AI RISK ENGINE

From Financial TEXTS to Measurable Risk

S&P Global × CRISIL Campus Hackathon 2026
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Sentiment alone cannot quantify exposure, losses or capital impact.



01

News + social posts

Credit deterioration and market-stress signals

02

Financial NLP engine

Traceable, structured intelligence

03

Financial consequences

A · Portfolio rebalancing
B · Bank stress testing

NOVELTY · One connected workflow from unstructured text to portfolio-level and bank-level risk analysis.

Made with GAMMA

Product architecture: from data sources to risk dashboard



Separate financial-data path

Prices / market data · SEC EDGAR XBRL debt
Ratings + evidence URLs · FRED / crisis data
Hypothetical loans, bonds and hedges

→ Data flow · → API communication · ⇢ Risk calculations and outputs

Live impact is formula-based

Learned impact model not wired into the engine.
Historical inputs enter A and B separately;
NLP does not generate market or portfolio data.

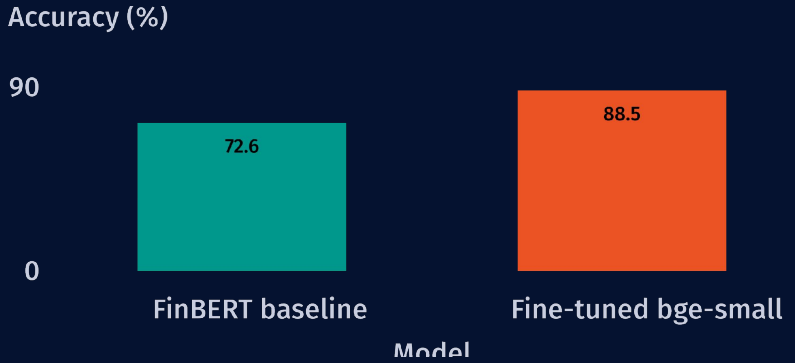
NOVELTY · Shared intelligence, two risk domains. Not production-certified; stress outputs not independently validated.

Made with GAMMA

Measured NLP performance: sentiment and event models

Text → sentiment → event classification → impact scoring → credit-specific alert validation

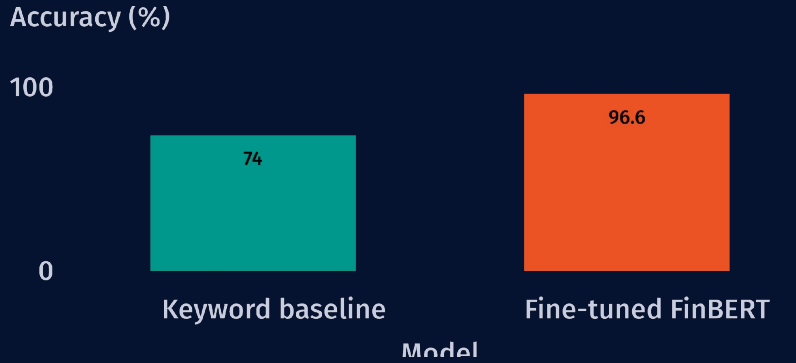
Sentiment · TFNS test set · 2,388 tweets



Accuracy +15.9 points · macro-F1 0.669 → 0.848

CPU 13.5 → 48 texts/s · size 438 → 133 MB

Events · TFN-topic · 4,117 tweets



Four mapped classes · macro-F1 0.505 → 0.933

Default, Credit Downgrade and Covenant Breach still need independent labelled validation.

Live impact is formula-based. Learned impact missed its original correlation target; no headline accuracy or credit-event performance claim.

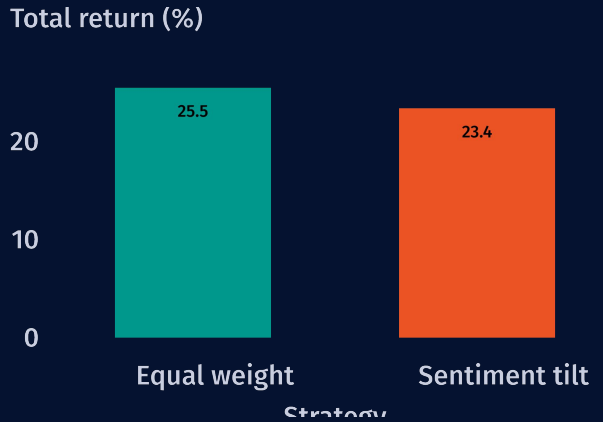
NOVELTY · Compact sentiment model, separate event classifier and credit-specific validation in one pipeline.

Made with GAMMA

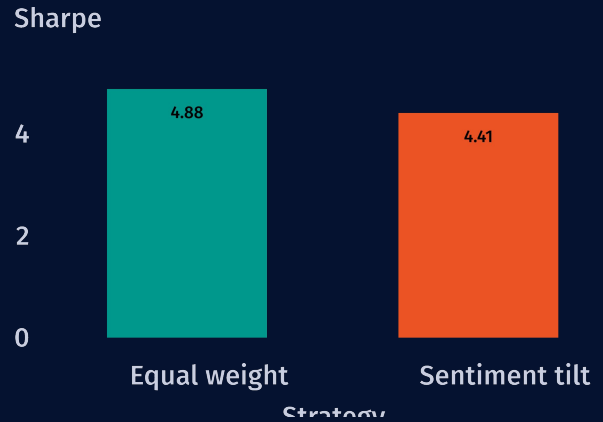
Module A: signals into constrained portfolio decisions

Initial backtest · 9 April to 10 June 2020 · equal weight vs sentiment tilt

Total return (%)



Sharpe



CONSTRAINTS

15%

Max per stock · long-only

5 bp

Cost of turnover

Previous 3-trading-day sentiment mean; apply next trading day after the text's New York date.

Equal weight did better in this window. Sentiment tilt had about 10× higher turnover.

NOVELTY · Explicit constraints, costs and no-look-ahead timing, not a claim of superior returns.

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Module B: from a credit event to bank capital impact

15 companies · 85 positions · hypothetical bank book dated 31 December 2019
SEC EDGAR XBRL debt figures · S&P ratings with evidence URLs

01

Qualifying alert

Credit-specific gate

02

Historical scenario

GFC · Eurozone · COVID-19

03

Risk translation

Rating-based PD · spreads · correlation contagion

04

Revaluation

Loans · bonds · CDS · selected swaps

05

Capital impact

Losses · ECL · IFRS 9 · RWA · CET1 · 99% VaR / ES · reverse stress

SCREENSHOT PLACEHOLDER

Module B dashboard

CET1 before / after: [not supplied]
Loss values: [not supplied]

Validation exposes calibration limits

- 2 of 3 alert cases had correct loss direction; magnitudes about 18–75× too large.
- 12-crisis analogue error $45.7M vs $47.0M naive average; monthly forecast did not beat its naive baseline.

Assumed exposures, hedges, LGD and initial CET1 · static 2019 book vs some 2020 alerts · fixed rates · some hedge risks unmodelled · stress tests, not forecasts

NOVELTY · Credit alerts linked to portfolio revaluation and capital-adequacy analysis.

Made with GAMMA

Built as a reproducible financial-risk application

Data + calculations

Python · NumPy · pandas

NLP inference

PyTorch · Transformers
ONNX int8 where enabled

Service + interface

FastAPI · React · TypeScript

Reproducibility

Docker · automated testing / CI
Modular calculations with tests

SCREENSHOT PLACEHOLDER

Signal replay

Event → shared structured signal → downstream analysis

Insert an actual application capture

SCREENSHOT PLACEHOLDER

Module A / Module B results

API endpoints for signals, portfolio data and stress-test reports

No fabricated UI, latency or test counts

NOVELTY · NLP inference, event validation, portfolio calculations and scenario credit analysis share one application architecture.

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A connected, auditable financial risk workflow

01

End-to-end risk translation

02

Credit-specific alert gating

03

Constraint-aware portfolio decisions

04

Disclosure-grounded hypothetical bank portfolio

05

Scenario-based capital analysis

IMPLEMENTED

- Signals + credit gating
- Rebalancing + stress tests
- Dashboard + API

LIMITATIONS

- Hypothetical exposures / assumptions
- 2019 book vs 2020 alerts
- Unvalidated credit detection; uncalibrated losses

NEXT STEPS

- Label and validate credit events
- Improve impact generalization vs baselines; calibrate losses
- Extend rate, funding and hedge modelling; more out-of-sample tests

Financial NLP + constrained portfolio decisions + transparent bank stress testing

One connected and testable risk workflow.

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